

TrES-3b

R-2001

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130508 · created 2026-08-07T16:01:48+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456420.794 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.055
Transit depth (Rp/Rs)^2	3.86 +/- 2.15 [%]
Orbital Inclination (inc)	81.05 +/- 2.78
Airmass coefficient 1 (a1)	1.025 +/- 0.062
Airmass coefficient 2 (a2)	-0.01 +/- 0.033
Scatter in the residuals of the lightcurve fit is	6.06 %
Best Comparison Star	#3 - [481, 86]
Optimal Aperture	2.11
Optimal Annulus	5.17
Transit Duration (day)	0.044 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.055 vs 0.16309 (deviation 0.62σ)
Duration fit vs published	0.044 d vs 0.0567 d (deviation 0.44σ)
Mid-time O–C	-3.4 min
Depth SNR	3.6
Residual scatter	6.06 %

Light curve

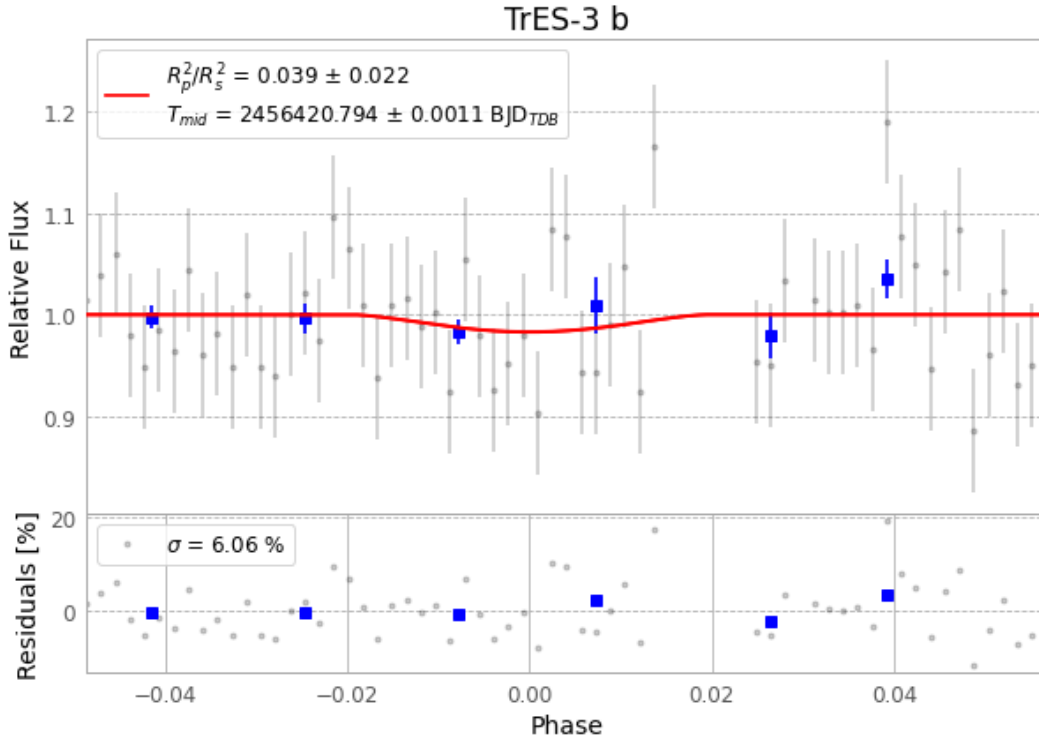


Figure 1 — FinalLightCurve_TrES-3 b_2013-05-07.png (SHA-256 b5e25c623e2a5b6c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [276, 195], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 52ef369881708cf1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

67 FITS frames (44 MB), TRES-3130508052712.FITS → TRES-3130508084512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130508.log (c0a705027cdf...), FinalLightCurve_TrES-3 b_2013-05-07.png (b5e25c623e2a...), FinalLightCurve_TrES-3 b_2013-05-07.csv (de5990d91017...)

Compute resources

Wall time 143.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 16:01Z.